

Time Series and ARIMA

DSC 340 Applied Machine Learning

EMC Ch. 8

The Plan

- Introduce time series
- ARIMA model
 - We are omitting many technical details
- Other methods

Time Series

- **Discrete time**

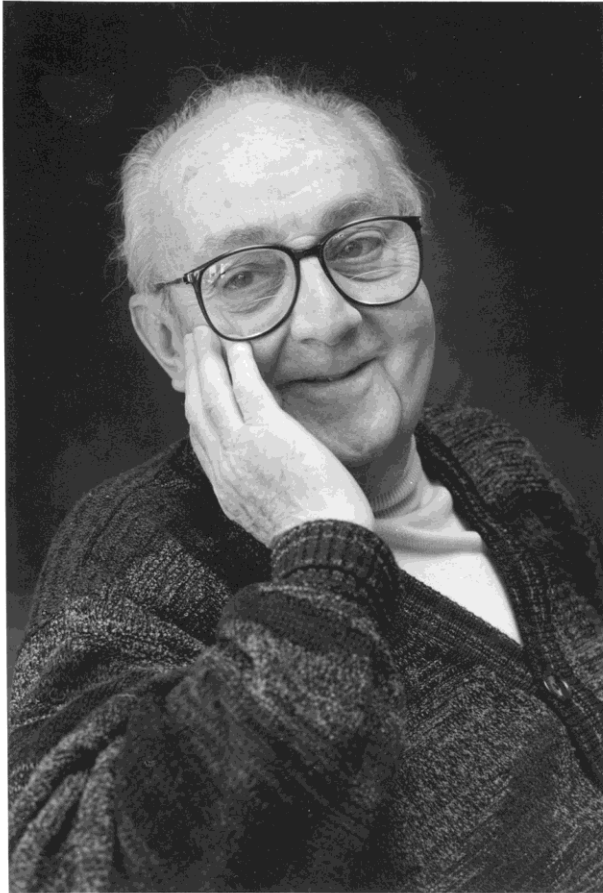
- Measurements are "evenly spaced" (time $t = 0, 1, 2, \dots$)
- Common examples: daily, monthly, every second
- **Usually y_t is NOT independent of y_{t-1}**
- To think about: **missing observations** (e.g., markets closed)

- **Objectives**

- **Understand** the pattern (**infer** and **model** the structure)
- **Forecast** (predict future values; extrapolating)

- **Applications**

- Retail sales, inventory management, stock trading

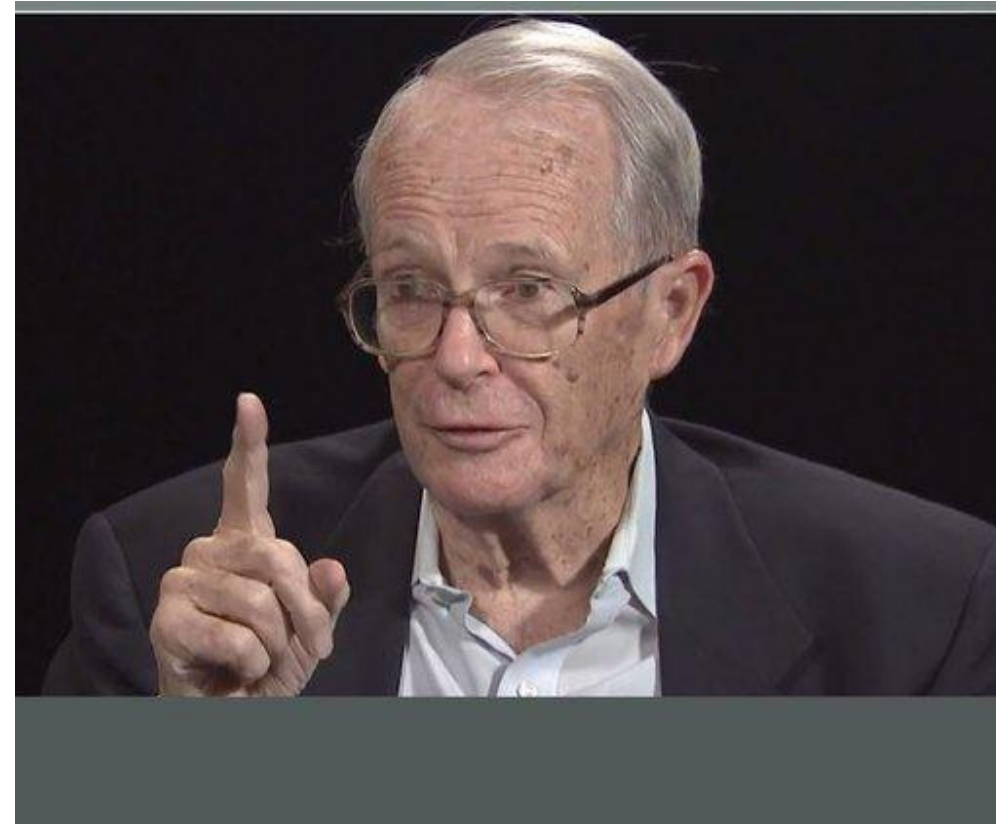


George Box

(Quote: "all models are wrong")

(1919-2013)

<https://commons.wikimedia.org/wiki/File:GeorgeEPBox.jpg>



Gwilym Jenkins

Welsh statistician and systems engineer

(1932-1982)

<https://alchetron.com/Gwilym-Jenkins>

Box-Jenkins Methodology

- **Trend**

- Long-term movement up or down ("secular trend")
- Example: real estate values gradually increase in some market

- **Seasonality**

- Examples: swimsuits and ski jackets; vacation rentals

- **Cyclic**

- Similar to seasonal, but not tied to specific dates
- Oil industry: "boom and bust"

- **Random**

- "White noise"; often assume normally distributed

Fitting with Box-Jenkins

1. Condition the data and select a model
 - Identify and account for trends or seasonality
 - Examine the adjusted time series and determine suitable model
2. Estimate model parameters (fit model)
3. Assess the model (go to step 1 if necessary)
 - Note that since a sequence, we can't use *randomized* train-test split
 - Common to partition into two *sequences* to test how well it will generalize



<https://otexts.com/fpp2/accuracy.html>

ARIMA

- AR **Auto**regressive (auto = self)
 - I Integrated (seasonal model)
 - MA Moving Average
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- Really a **hybrid** model (AR + MA = ARMA, also ARIMA)
 - Often incorporates several different models (not just these), e.g.,
 - Long-term trend
 - Seasonal patterns
 - Vectorized (multi features)

Assumptions

- Stationary
 - Consistent in the time under observation (e.g., probabilities don't change)
 - Often becomes stationary after correcting for long-term and seasonal trends
- Current events only depend on a fixed number of past events
 - Markov assumption
 - "Of order p " (or q , etc.)
- Error term ε_t
 - Usually assume $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ for all t
 - White noise process
 - Random, independent fluctuations

Autoregressive Model

- AR(p): Read as "autoregressive model of order p "

$$y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

- y_t is the value at time t
- ϕ_k is a constant for $k = 1, 2, \dots, p$
- δ is some constant (informally, baseline or average)
- $\phi_p \neq 0$
- $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ for all t

Moving Average Model

- MA(q): Read as "moving average model of order q "

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

- y_t is the value at time t
- θ_k is a constant for $k = 1, 2, \dots, q$
- $\theta_q \neq 0$
- $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ for all t

ARMA Model

- Combined model
- ARMA = AR(p) + MA(q)

$$y_t = \delta + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

Trends in MA models

- Stationary assumption assumes no trend
- Often there is a trend, so we have to "correct" for this
- One approach: Fit linear (or higher order model) then subtract off
 - You can imagine this as "tilting" the model CW or CCW to make it flat
- More common approach: Differencing

$$d_t = y_t - y_{t-1} \text{ for } t = 2, 3, \dots, n$$

- Higher-order differencing ("twice differenced" or of order d)
 - But be careful: similar to fitting higher-order polynomials!
 - Can unnecessarily increase the variance

ARIMA Model

- Autoregressive **integrated** moving average model

$$\text{ARIMA}(p, d, q) \times (P, D, Q)_s$$

- p, d, q same as before AR(p), MA(q), d is # of times differenced
- s denotes seasonal period (e.g., 365 if annual, 7 if weekly, etc.)
- P # of terms in AR model across the s periods
- D # of differences applied across the s periods
- Q # of terms in the MA model across the s periods

Evaluating model

- Metrics such as AIC, AICc, BIC
- Normality and constant variance
 - Is the error actually normally distributed?
- How well does it extrapolate?
 - Example: train model for first 10 months
 - Evaluate model on the next 2 months of historical data
 - **Again: CAUTION! Extrapolation assumes stationarity!**
 - Long-term projections are particularly risky

Other considerations

- Not a causal model
 - Doesn't tell us how to improve our sales!
- Not resilient to severe shocks
- Most useful for short-term forecasting

Parameters and Hyperparameters

- What are the **parameters** and **hyperparameters**?
- Hyperparameters include: p, d, q
 - You also need to decide on timestep and seasonal period
 - But you probably make that decision even earlier, when preparing the data
- Parameters may include:
 - Coefficients and intercept for AR: $\phi_1, \dots, \phi_p$ and δ
 - Coefficients for MA: $\theta_1, \dots, \theta_q$
 - Differencing coefficients: $d_2, \dots, d_n$
 - There may be others: Remember, these are often complex, integrated models

Implementation

- Supported by R (see book)
- Also Jupyter/sklearn (see tutorial linked from Moodle)
- Many custom toolkits for stock trading, etc.

Other Methods

- Multivariate time series analysis (VARIMA = "vector ARIMA")
- Markov chains, hidden Markov models (HMMs), Kalman filters (state)
- Recurrent neural networks (RNNs): multi-layer perceptron with cycles
- Several other methods (see book, p. 253)